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SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0502- COMPUTER VISON

MODEL LAB PRACTICALS

SET 9

1. Implement a face detection algorithm using Open CV to detect and locate human faces in the images.
2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.
3. Implement a function to reverse the frames of the video to create a video in reverse mode using Open CV.
4. Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

Viva attended

93/100

SET - 9

Model Practical

Name : B. Srimathi
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Course code : ITA0502
Course name : Computer vision
Slot : C

1 ## Face Detection Algorithm

Aim:

⇒ To detect and locate human faces in an image using opencv.

Procedure:

=> Load the source image.
=> Convert it to grayscale.
=> Load the Haar Cascade classifier.
=> Detect face coordinates
=> Draw bounding rectangles
=> Display the final output.

Program:

```
import cv2
img = cv2.imread('face.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
faces = face_cascade.detectMultiScale(gray, 1.11, 4)
for (x,y,w,h) in faces:
    cv2.rectangle(img, (x,y), (x+w, y+h), (0, 255, 0), 2)
cv2.imwrite("result.jpg", img)
```

Input :

⇒ A standard jpg image containing one or more human faces

Output

⇒ The image is successfully processed and green bounding boxes faces are accurately and detected faces.

Result:

⇒ Face succesifully detected and highligted with a green bounding box.

2. Image Handling and Erosion

Aim:

⇒ To read an image and apply morphological erosion using Opencv.

Procedure:

⇒ Read the input image

⇒ Define a structuring kernel.

⇒ Apply the erosion function.

⇒ Display original and erode images.

Program:

```
import cv2
import numpy as np
img = cv2.imread('binary.jpg', 0)
kernel = np.ones((5,5), np.uint8)
eroded = cv2.erode(img, kernel, iterations=1)
cv2.imwrite('eroded.jpg', eroded)
```

Input:

⇒ A binary (black and white) image file named in binary.jpg by using grayscale image

Output:

⇒ A new image saved as eroded.jpg

Result:

⇒ Object boundaries are successfully thinned out, reducing noise.

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3. Video Frame Reversal

Aim:

⇒ To reverse the frames of a video to create a video in reverse mode using Open cv.

Procedure:

⇒ Open the video file.
⇒ Read frames into the list
⇒ Get video
⇒ Initialize video writer object
⇒ Write frames in reverse order

=> Save the output file

=> End

Program:

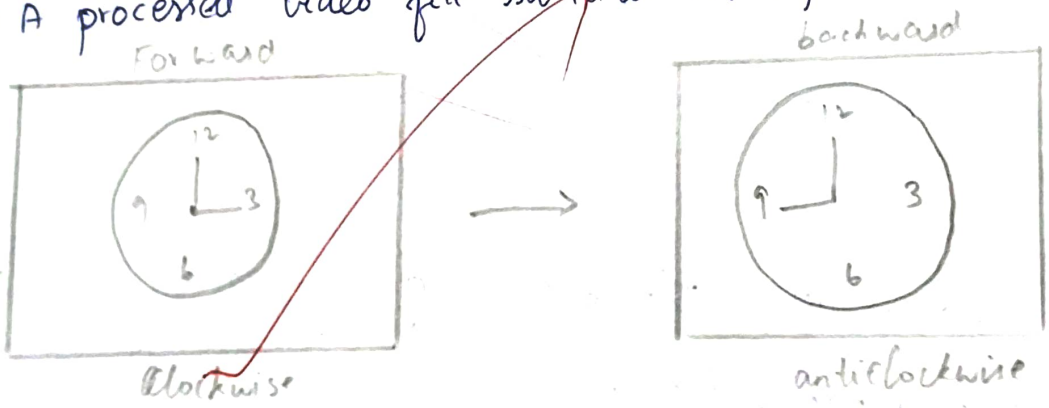
```
import cv2
cap = cv2.videoCapture('video.mp4')
frames = []
while cap.isopened():
    ret, frame = cap.read()
    if not ret: break
    frames.append(frame)
cap.release
out = cv2.videowriter('rev.mp4')
out.release()
```

Input:

=> A standard video file named video.mp4.

Output:

=> A processed video file saved as rev.mp4



Result:

Video frames are successfully saved in reverse sequence, creating a backward playing video.

Image Cropping, Copying and Pasting

Aim:

⇒ To implement an Image cropping, copying and pasting to select a region of interest from the source image using open CV.

Procedure:

⇒ Read the input image.
⇒ Define ROI coordinate bounds
⇒ Crop using Num-Py slicing
⇒ Define target destination coordinates
⇒ Paste cropped ROI back
⇒ Save the modified image

Program:

```
import cv2
img = cv2.imread('source.jpg')
roi = img[100:250, 200:350] # crop
img[300:450, 400:500] = roi # paste
cv2.imwrite('roi_output.jpg', img)
```

Input:

⇒ The image file named source.jpg

Output:

⇒ The output image is saved as roi_output.jpg

Cropping Copying pasting

Result:

-> The specified Region of interest (ROI) is cropped from its original position and pasted onto another part of image.